



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Assignment 1 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BCE		
Semester:	7 th	Batch:	FA22	Section:	A, B
Submission Date:	29 th September, 2025	Maximum Marks:		Date Given:	24 th September, 2025
Name:		Registration Number:			

Important Instructions / Guidelines:

- Do your own work, PLAGARISM will be graded as ZERO
- No late submission.

Question 1:

[CLO1-PLO3-C5]

[10 Marks]

Draw a state diagram and **design** a controller for an FSM with a 2-bit input (AB) and an output (Y). The FSM transitions are defined as follows:

- When AB changes from 00 to 01, Y is asserted ($Y = 1$) for one clock cycle, then returns to 0, even if AB remains 01.
- When AB changes from 01 to 10, Y is asserted for two clock cycles, then returns to 0, even if AB remains 10.
- When AB changes from 10 to 11, Y is asserted for three clock cycles, then returns to 0, even if AB remains 11.

Note: The input (AB) follows the sequence $00 \rightarrow 01 \rightarrow 10 \rightarrow 11$. If any undesired input occurs, the FSM resets itself and returns to its initial state.